

RESEARCH PROPOSAL TEMPLATE

Research title <i>(English)</i>	Reliable IoT Sensor Data Transmission using Hamming Code (7,4)
Sub-committee	Information Technology (IT)
Group name	SE1940-G05
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Abstract

This study proposes an end-to-end simulation framework for Reliable IoT Sensor Data Transmission utilizing Hamming Code (7,4) to resolve the challenge of data corruption over noisy wireless channels. In Internet of Things (IoT) ecosystems, continuous sensor telemetry is highly vulnerable to ambient signal interference and hardware noise, which frequently cause critical single-bit flips, severely undermining the reliability of real-time monitoring applications. To address this problem without the bandwidth overhead of traditional automatic repeat request (ARQ) retransmission mechanisms, this research implements a hardware-less software-based mathematical solution.

The methodology involves partition-slicing digitized continuous telemetry streams into standardized 4-bit data words (where $m = 4$ data bits). Using the optimal boundary condition where 2^r must be greater than or equal to $m + r + 1$, exactly 3 parity bits (where $r = 3$) are calculated via localized Exclusive-OR (XOR) operations and mapped into structured 7-bit codewords. Upon reaching the receiver gateway, a 3-bit Syndrome vector is computed to mathematically isolate the exact index of any corrupted bit, performing Single Error Correction (SEC) automatically.

The primary research objective is to validate the mathematical precision and computational efficiency of the Hamming mechanism across multiple fault-injection test scenarios. Evaluation metrics focus on

analyzing the Single-bit Error Correction success rate, measuring processing latencies, and verifying JSON log generation schema. The expected outcomes include a modular software library for error-correction codecs and verified performance logs demonstrating guaranteed data integrity, establishing a scalable foundation for lightweight, error-resilient communication protocols in resource-constrained IoT nodes.

Key words: Hamming Code (7,4), IoT Reliability, Error Correction, Sensor Telemetry, Signal Noise, Single Error Correction (SEC).

Introduction

1.1. Literature review

Ensuring data integrity in wireless Internet of Things (IoT) network transmissions remains a major challenge. Signal interference and hardware noise frequently cause bit-flip corruptions in transmitted sensor telemetry. Traditional solutions like Checksum and Cyclic Redundancy Checks (CRC) are highly effective at detecting errors but completely lack self-correction capabilities. Consequently, these protocols must rely on Automatic Repeat Request (ARQ) mechanisms to trigger data retransmissions.

However, recent academic studies demonstrate that constant ARQ retransmissions introduce substantial inefficiencies. They cause significant processing latencies, waste network bandwidth, and rapidly deplete the battery lifespans of resource-constrained IoT sensors. To bypass these limitations, Forward Error Correction (FEC) schemes have become a vital alternative by enabling instant error correction at the receiver side without retransmission overhead.

Nevertheless, implementing high-level FEC codes (such as Reed-Solomon or LDPC) directly on ultra-low-power microcontrollers creates a major technical contradiction due to their excessive computational and memory demands. This practical gap emphasizes the strict necessity for lightweight algorithmic frameworks. While historic literature mostly implements Hamming Code within physical hardware chips, there is a distinct shortage of localized, software-defined frameworks tailored for continuous IoT sensor streams. Adopting a modular, software-based Hamming Code (7,4) protocol provides an ideal compromise. It guarantees robust Single Error Correction (SEC) through simple logical XOR operations, maximizing data reliability while fully respecting the strict hardware and power limitations of edge IoT gateways.

1.2. The necessity of the research

This research addresses a critical technical problem: the high vulnerability of Internet of Things (IoT) sensor telemetry to bit-flip errors during wireless transmission, and the severe inefficiencies caused by traditional recovery methods.

In low-power wide-area networks (LPWAN) or radio frequency deployments, signal noise regularly corrupts sensor data. While traditional communication frameworks apply error-detection checks, they must rely on Automatic Repeat Request (ARQ) retransmissions to correct those errors. This approach creates a severe practical conflict. For resource-constrained IoT systems, constantly re-transmitting data rapidly depletes the edge sensor batteries, creates severe network latency, and wastes valuable bandwidth.

The originality and scientific significance of this study lie in developing a localized, software-defined end-to-end simulation framework that applies Hamming Code (7,4) directly to the continuous telemetry pipeline without relying on dedicated hardware. This software-based Forward Error Correction (FEC) solution allows the receiver gateway to automatically perform Single Error Correction (SEC) instantly at the point of reception. By eliminating the network overhead of ARQ retransmissions, this research provides a vital, lightweight mechanism to maximize data reliability, optimize power efficiency, and extend the operational lifecycle of low-power IoT networks.

1.3. Feasibility of Research

1. Analyze the feasibility of research.

Research Question: *This study focuses on a core research question: How can a software-defined Hamming Code (7,4) pipeline be designed and optimized to guarantee single-bit error correction in continuous IoT sensor streams without introducing hardware overhead or communication latency?*

The study aims to build a lightweight, logical error-correction framework that operates efficiently within the strict memory constraints of low-power IoT networks, ensuring data integrity at the receiver side.

Research Design: *The proposed study implements an experimental, software-based simulation design. The framework models an end-to-end telemetry pipeline divided into three main operational phases:*

- *The Encoding Phase: Takes incoming 4-bit data blocks, computes check values using logical Exclusive-OR (XOR) operations, and structures them into secure 7-bit transmission packages.*

- *The Channel Noise Phase: Injects controlled single-bit error variations into the stream to simulate real-world wireless signal interference.*
- *The Decoding Phase: Evaluates the incoming packages by computing a mathematical indicator called a Syndrome vector. This vector isolates the exact location of any flipped bit and fixes it automatically.*

Sample Size and Population: *Because this study focuses on a software-defined simulation rather than a physical hardware deployment, the study population consists of an infinite stream of simulated binary telemetry records. For the validation phase, the sample size is defined as a structured validation suite consisting of 1,000 randomized 4-bit sensory data iterations. These iterations are systematically tested across all 7 discrete single-bit error positions to verify correction capabilities.*

Data Sources: *The primary data sources are dynamically simulated digital sensor bitstreams. These streams mimic continuous real-world environmental monitoring systems, generating typical telemetry readouts such as temperature, humidity, and vibration frequencies. Since the focus is on raw binary error correction, no external physical hardware or external historical datasets are required.*

Budget and Funding: *The total estimated budget for conducting this study is \$0.00. The project is entirely software-driven and relies exclusively on open-source programming languages, free code editors, and built-in software development testing suites. No external funding sources are required.*

Timeframe: *The study is structured over a concise 5-week execution schedule:*

- *Weeks 1-2: Domain exploration, architectural design, literature review, and keyword selection.*
- *Week 3: Core software implementation of the Encoder and Decoder logical algorithms.*
- *Week 4: Unit test execution, fault-injection tests, and performance metric evaluation.*
- *Week 5: Final compilation of the research paper draft and project package assembly.*

Potential Challenges and Risks: *The primary technical risk is the occurrence of multiple-bit errors (2 or more bits flipped within a single block), which exceeds the standard capacity of the Hamming Single Error Correction (SEC) protocol. This challenge is addressed by designing a robust system status code framework. When an uncorrectable multi-bit error is detected, the system flags it under status code "10", gracefully bypassing corruption and triggering a simulated repeat-request warning to protect system stability.*

2. Research Objectives

The study aims to achieve the following clear objectives:

- 1. To design and implement a highly efficient, modular software library containing the logical components for a Hamming Code (7,4) Encoder and Decoder.*
 - 2. To build a localized fault-injection simulation engine that accurately models wireless communication signal degradation and single-bit flips.*
 - 3. To establish a standardized data log structure utilizing a custom JSON output schema to track communication health metrics and error telemetry in real time.*
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3. Research Scope

Inclusions: *The study strictly covers the mathematical verification of the Hamming boundary condition, parity check distributions, and automated Single Error Correction (SEC) algorithms. The scope is specifically tailored for processing segmented 4-bit data words into 7-bit codewords.*

Exclusions: *Multiple-bit error corrections (MEC) are excluded from this framework. Physical microcontroller deployment (such as hardware interfacing on Arduino or Raspberry Pi) is omitted; the research acts as a high-fidelity software architecture model.*

4. Approach and Method

The research utilizes an Even-Parity validation methodology driven by logical XOR operations. The raw data stream is segmented into 4-bit words represented as $D = \{D1, D2, D3, D4\}$. To provide self-repair capabilities, 3 parity bits ($P1, P2, P4$) are embedded into positions representing powers of 2.

The data structure and bit mapping follow a strict 7-bit layout:

- Position 1: $P1$ (Checks positions 1, 3, 5, 7)*
- Position 2: $P2$ (Checks positions 2, 3, 6, 7)*
- Position 3: $D1$*
- Position 4: $P4$ (Checks positions 4, 5, 6, 7)*
- Position 5: $D2$*

- Position 6: $D3$
- Position 7: $D4$

Mathematical Equations:

1. Encoding Stage:

- $P1 = D1 \text{ XOR } D2 \text{ XOR } D4$
- $P2 = D1 \text{ XOR } D3 \text{ XOR } D4$
- $P4 = D2 \text{ XOR } D3 \text{ XOR } D4$
- Complete Codeword Layout: $C = \{P1, P2, D1, P4, D2, D3, D4\}$

2. Decoding Stage (Syndrome Generation):

- $S1 = P1' \text{ XOR } D1' \text{ XOR } D2' \text{ XOR } D4'$
- $S2 = P2' \text{ XOR } D1' \text{ XOR } D3' \text{ XOR } D4'$
- $S4 = P4' \text{ XOR } D2' \text{ XOR } D3' \text{ XOR } D4'$

The binary combination $[S4, S2, S1]$ forms a decimal value from 1 to 7. If the decimal value is 0, the data is clean. Any non-zero value directly pinpoints the exact bit index that has been corrupted, allowing the decoder to automatically invert the bit back to its valid state.

5. Research plan

Describe the research plan (Give a concise description of each task, and their expected results, schedule, and person in charge as in the following table).

No.	Date	Task	Output	Person in charge
1	11/05/2026 – 24/05/2026	Study the topic and design data input/output format.	Documentation files	Huynh Anh Khoi, Nguyen Nhat An

2	11/05/2026 – 24/05/2026	Read related papers and collect keywords.	Paper list and matrix	All Members
3	25/05/2026 – 31/05/2026	Write code for Encoder and Decoder logic.	Source code files	Huỳnh Thị Thanh Nguyên, Huynh Anh Khoi
4	01/06/2026 – 07/06/2026	Create error test cases and check code coverage.	Test suites and JSON schema	Nguyen Duong Hoai An, Nguyen Nhat An
5	08/06/2026 – 14/06/2026	Run experiments, make charts, and write final report.	Final paper draft	All Members

Expected Results:

- *Theoretical implications: Prove that Hamming Code (7,4) works correctly on software without needing hardware.*
- *Methodological implications: Provide a simple framework to test and inject errors into data streams.*
- *Practical implications: Create a lightweight library that fixes 1-bit errors, saves network bandwidth, and reduces sensor battery cost.*
- *Plans to apply this research in practical: Use this code in IoT gateways to clean sensor data before sending it to the cloud server.*

References:

- [1] R. W. Hamming, "Error detecting and error correcting codes," *Bell System Technical Journal*, vol. 29, no. 2, pp. 147-160, Apr. 1950.
- [2] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, and E. Cayirci, "A survey on sensor networks," *IEEE Communications Magazine*, vol. 40, no. 8, pp. 102-114, Aug. 2002.

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COMMITMENT ON RESPONSIBLE ARTIFICIAL INTELLIGENCE USAGE IN RESEARCH

We are authors of the research paper titled **“Reliable IoT Sensor Data Transmission using Hamming Code (7,4)”**, would like to ensure that all members comply with the rules relevant to Artificial Intelligence (AI) regulations and ethical guidelines in research practices, which are described in details as follows:

1. AI is only used as assistant in correcting grammar, structure and phrasing in academic writing;
2. AI usage should NOT be engaged with the creation of content or drafting the research paper, production of fake data or literature synthesis;
3. AI usage should be checked and confirmed by all authors. Each author shall bear full responsibility in the accuracy, transparency and originality of their sections, including citations and reference;
4. All authors should disclose the use of generative AI and AI-assisted technologies to other co-authors, helping them understand where and how AI was applied in research preparation.

Note:

- *Research proposal must be presented in English/Japanese, A4 paper, single column, Times New Roman 11, and a maximum length of 10 pages (excluding references and appendices)*
- *For citations and references, please refer to:*

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